

The exact Hall ratio and extremizers of the sixth Mycielski graph

5 October 2026

Abstract

We give a computer-assisted determination of the Hall ratio of the 47-vertex Mycielski graph, with the convention $M_2 = K_2$ and $M_{r+1} = \mu(M_r)$. Its Hall ratio is $10/3$. Exactly 1,990 induced vertex subsets attain this value, forming 199 orbits under the full ambient automorphism group, each of size ten. Every maximizer has twenty vertices, independence number six, and contains the final apex. The proof uses 857 maximal-independent-set constraints, obtained independently by a recursive neighborhood description and by standard direct enumeration. Exact rational certificates establish the upper bound and equality reductions; two different exact algorithms independently enumerate all extremizers. The main results are the exact finite Hall value and complete census. The recursive description organizes the certificate data; no general algorithmic efficiency improvement is claimed.

1 The finite result and its scope

For a nonempty finite graph H , its Hall ratio is

$$\rho(H) = \max_{\emptyset \neq S \subseteq V(H)} \frac{|S|}{\alpha(H[S])}.$$

Here α denotes the independence number, and $H[S]$ is the induced subgraph. We abbreviate $\alpha(H[S])$ to $\alpha(S)$ when the ambient graph is clear. We use the ordinary Mycielski construction: $\mu(G)$ has original vertices $V(G)$, an independent clone layer $V(G)'$, and an apex z . Original vertices retain their edges; u is adjacent to u' precisely when $uv \in E(G)$; and z is adjacent to every clone. Set $M_2 = K_2$ and $M_{r+1} = \mu(M_r)$, so the successive orders are 2, 5, 11, 23, 47.

Theorem 1.1 (Computer-assisted finite classification). *The graph M_6 satisfies $\rho(M_6) = 10/3$. Every maximizing subset has order 20, independence number 6, and contains the final apex. There are exactly 1,990 labelled maximizing subsets, in 199 orbits under $\text{Aut}(M_6) \cong D_5$. Every orbit has size 10. The final original/clone/apex layer counts are as follows.*

<i>Originals</i>	<i>Clones</i>	<i>Apex</i>	<i>Orbits</i>	<i>Labelled subsets</i>
13	6	1	109	1,090
14	5	1	83	830
15	4	1	7	70
<i>Total</i>			199	1,990

The orbits in this theorem concern embedded subsets under automorphisms of the ambient graph. They are not asserted to be distinct abstract isomorphism types of the induced graphs.

Cropper, Gyárfás and Lehel [1] give the values $\rho(M_2) = 2$, $\rho(M_3) = 5/2$, $\rho(M_4) = 8/3$ and $\rho(M_5) = 3$, and prove that the sequence of Hall ratios is unbounded. The cardinality-profile formulation used below predates this calculation; see Barnett [2]. Steiner [3] records that boundedness of $\chi_f(M_r)/\rho(M_r)$ remains open. The present finite calculation does not resolve that asymptotic question. The main results recorded here are the exact finite Hall value and extremizer census. The structural identities are convenient proof lemmas, possibly already known; no first-proof claim is made for them, and publication priority for the finite results has not been established.

2 A recursive description of the certificate constraints

Throughout this section, G is a finite simple graph with $n \geq 1$ vertices. Let $\mathcal{I}(G)$ contain all independent sets, including the empty set, and write $i(G) = |\mathcal{I}(G)|$ and $\text{mis}(G)$ for the number of maximal independent sets. For $I \subseteq V(G)$, $N(I)$ is the union of its open neighborhoods. Define

$$\mathcal{D}(G) = \{N(I) : I \in \mathcal{I}(G)\}, \quad d(G) = |\mathcal{D}(G)|.$$

Theorem 2.1 (Neighborhood-union identity). *For every nonempty finite simple graph G ,*

$$\text{mis}(\mu(G)) = \text{mis}(G) + d(G). \quad (1)$$

For $N \in \mathcal{D}(G)$ put $J_N = \{v : N(v) \subseteq N\}$. The maximal independent sets avoiding the apex are exactly

$$K_N = J_N \cup (V(G) \setminus N)', \quad N \in \mathcal{D}(G),$$

with one distinct set for each N . Those containing the apex are exactly $\{z\} \cup I$, where I is maximal independent in G .

Proof. Choose independent I with $N = N(I)$. Then $I \subseteq J_N$ and $J_N \cap N = \emptyset$. Indeed, if $v \in J_N \cap N$, it has a neighbor $i \in I$, and $N(v) \subseteq N$ would force $i \in N$, contrary to the independence of I . It follows that J_N is independent and $N(J_N) = N(I) = N$.

Thus K_N is independent. An excluded original vertex has a neighbor among its selected clones, by the definition of J_N . An excluded clone has a neighbor in $I \subseteq J_N$. There is a selected clone: $V \setminus N$ contains I when I is nonempty, and equals V otherwise. Hence the apex also has a neighbor in K_N , so K_N is maximal.

Conversely, a maximal independent set avoiding z , with original part I , must contain every available clone, namely $(V \setminus N(I))'$. Maximality then forces its original part to be $J_{N(I)}$. The clone part uniquely determines N . A maximal independent set containing z contains no clone, and its original part is maximal in G . These two bijections prove (1). $\square$

Corollary 2.2. *All maximal independent sets of $\mu(G)$ can be generated in $O(n^2 i(G))$ elementary Boolean operations, with $O(n^2 + ni(G))$ bits of working and output storage. Their number is at most $2i(G)$.*

Proof. Enumerate independent sets of G , collect their distinct neighborhood unions, and identify maximal sets by $I \cup N(I) = V(G)$. Construct the sets in Theorem 2.1. Boolean adjacency arrays give the stated bounds; bitsets improve the implementation. Finally, $d(G) \leq i(G)$ and $\text{mis}(G) \leq i(G)$. $\square$

Remark 2.3 (Comparison with standard enumeration). Tsukiyama, Ide, Ariyoshi and Shirakawa [4] already enumerate all maximal independent sets of a graph with N vertices, M edges and K outputs in $O(NMK)$ time and $O(N + M)$ space. Corollary 2.2 is not an output-sensitive improvement. For an edgeless base on $n \geq 1$ vertices, $i(G) = 2^n$, while $\mu(G)$ consists of n isolated originals and $K_{1,n}$, and has exactly two maximal independent sets. The recursive implementation visits 2^n base sets; the standard enumeration bound on this graph is $O(n^2)$. The identity is therefore used only as a convenient recursive generator of certificate data. The independent M_6 audit also uses direct Bron–Kerbosch enumeration on the complement, without this identity. The standard bound concerns row generation, not the Hall binary optimization or the full extremizer census. No polynomial-time Hall optimization or small-certificate claim follows from the generator bound.

For the present instance, the recursive description gives

$$i(M_5) = 7,407, \quad d(M_5) = 778, \quad \text{mis}(M_5) = 79, \quad \text{mis}(M_6) = 857. \quad (2)$$

Proposition 2.4 (Induced-subset recurrence). *Let $A, B \subseteq V(G)$ and $\varepsilon \in \{0, 1\}$. For the induced subset consisting of originals A , clones B' , and the apex when $\varepsilon = 1$, its independence number equals*

$$\max \left\{ \varepsilon + \alpha(G[A]), \max_{\substack{I \subseteq A \\ I \in \mathcal{I}(G)}} (|I| + |B \setminus N(I)|) \right\}. \quad (3)$$

Proof. An independent set avoiding the apex chooses independent original part I and may include every available clone. A set using the apex chooses no clone and may include a maximum independent set in $G[A]$. If $\varepsilon = 0$, the first term is redundant and does not change the maximum. $\square$

The same decomposition gives the exact independence-polynomial identity

$$Z_{\mu(G)}(t) = tZ_G(t) + \sum_{I \in \mathcal{I}(G)} t^{|I|} (1+t)^{n-|N(I)|}, \quad Z_G(t) = \sum_{I \in \mathcal{I}(G)} t^{|I|}.$$

For example, it yields $i(M_6) = 39,473,983$ without enumerating those sets.

3 An exact certificate for the Hall ratio

Let $K_1, \dots, K_q$ be all maximal independent sets of a graph H , and let $x_v \in \{0, 1\}$ indicate a subset S . Then

$$\alpha(H[S]) \leq k \iff \sum_{v \in K_j} x_v \leq k \quad (1 \leq j \leq q). \quad (4)$$

Every independent set extends to a maximal one, which proves the reverse implication; each intersection $S \cap K_j$ proves the forward one. Thus maximizing $\sum_v x_v$ under (4) is an exact binary model for the cardinality profile. For M_6 it has 47 variables and 857 independent-set rows.

3.1 Rational leaves and binary splits

At a certificate node, fix vertices O to one and Z to zero, leaving $U = V(H) \setminus (O \cup Z)$ free. Set $b_j = k - |K_j \cap O|$. A rational-dual leaf supplies $y_j \geq 0$ and $z_v \geq 0$ satisfying

$$\sum_{j: v \in K_j} y_j + z_v \geq 1 \quad (v \in U). \quad (5)$$

Multiplying the residual rows by y_j and the bounds $x_v \leq 1$ by z_v proves

$$|S| \leq B := |O| + \sum_j y_j b_j + \sum_{v \in U} z_v. \quad (6)$$

If the exact rational B is below an excluded integer cardinality T , that node contains no subset of order at least T . An internal node splits on a free binary variable and checks both assignments. The certificate therefore covers every subset. Checking the displayed inequalities uses exact integer and rational arithmetic; numerical optimization is used only to propose weights or branches. Once the complete independent-set constraint system is established separately, the rational branch certificate can be checked in time polynomial in the combined explicit input and certificate bit length. This statement does not certify completeness of an arbitrary supplied constraint list.

3.2 Upper bound and witness

For $k \in \{1, 3, 4, \dots, 14\}$, the supplied certificate excludes

$$T_k = \left\lfloor \frac{10k}{3} \right\rfloor + 1.$$

The case $k = 5$ uses 130 splits and 131 rational leaves. Each of the other twelve cases uses one leaf, for 273 nodes in total. The missing case $k = 2$ is elementary: every six vertices of a triangle-free graph contain an independent triple. Given a vertex, three neighbors are independent; otherwise it has three nonneighbors, among which a nonedge together with the vertex gives an independent triple. Thus $\alpha \leq 2$ implies $|S| \leq 5$. For $k \geq 15$, use $|S|/k \leq 47/15 < 10/3$. The graph is checked to be triangle-free, so these cases prove $\rho(M_6) \leq 10/3$.

For the reverse bound, label originals $0, \dots, n-1$, their clones $n, \dots, 2n-1$, and the apex $2n$ at each step. The subset

$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 11, 12, 15, 19, 21, 22, \\ 26, 32, 33, 41, 45, 46\}$$

has twenty vertices and independence number six. Its maximum intersection with all 857 maximal independent sets is exactly six; $\{0, 2, 5, 7, 11, 41\}$ is one independent six-set. The independent audit also checks its independence number by a direct include/exclude recursion on these twenty vertices. Hence $\rho(M_6) = 10/3$.

4 Classifying all equality subsets

Equality forces $\alpha(S)$ to be divisible by three. The only possible pairs are

$$(10, 3), \quad (20, 6), \quad (30, 9), \quad (40, 12).$$

Additional exact certificates exclude the first, third and fourth pairs using 19, 1 and 1 nodes, respectively. One further leaf excludes a twenty-set with independence number at most six when the final apex is absent. These 22 nodes have nine binary splits and thirteen rational leaves.

Consequently every maximizing subset has twenty vertices, independence number six, and contains the apex. Write it as $A \cup B' \cup \{z\}$. The apex implies $\alpha(M_5[A]) \leq 5$, while $|B| \leq 6$. Exact base independence values give

$$\max_{\alpha(M_5[A]) \leq 4} |A| = 12, \quad \max_{\alpha(M_5[A]) \leq 5} |A| = 15. \quad (7)$$

These bounds are reverified by both complete enumerations. Since $|A| + |B| = 19$, they imply

$$|A| \in \{13, 14, 15\}, \quad \alpha(M_5[A]) = 5, \quad |B| = 19 - |A|.$$

By Proposition 2.4, such a pair is a maximizer exactly when

$$|B \setminus N(I)| \leq 6 - |I| \quad \text{for every independent } I \subseteq A. \quad (8)$$

This is an exact membership criterion within three layer classes. The orbit counts come from the complete finite enumeration below; no shorter structural derivation of 109, 83 and 7 is asserted.

4.1 The full ambient automorphism group

The graph M_3 is C_5 , in cyclic order $(0, 1, 2, 4, 3)$. At each later stage the apex is the unique maximum-degree vertex. Indeed, for a base of order n and maximum degree $(n - 1)/2$, the new original degrees are at most $n - 1$, clone degrees at most $(n + 1)/2$, and apex degree n . The property starts with C_5 and inducts.

Every automorphism therefore fixes the apex and preserves both layers. Its restriction to originals is an automorphism of the base. The bases are open-neighborhood twin-free, beginning with C_5 , and this forces the clone permutation to agree with the original permutation. Conversely, every base automorphism extends. Thus $\text{Aut}(M_6) \cong D_5$ has order ten. The independent checker obtains the ten automorphisms of the initial cycle by testing its 120 permutations, then lifts them and verifies adjacency, closure, and the unique-apex and twin-free hypotheses.

4.2 Two complete exact enumerations

The primary enumeration computes $\alpha(M_5[A])$ for all 2^{23} base subsets by vertex-deletion dynamic programming. It keeps only canonical A of sizes 13–15 with independence number five. There are 4,370 such D_5 representatives. For each one, choose an independent five-set $I_0 \subseteq A$ minimizing $|N(I_0)|$. Equation (8) requires at most one clone index outside $N(I_0)$. Thus, for $b = 19 - |A|$, enumerate only

1. b indices from $N(I_0)$;
2. $b - 1$ indices from $N(I_0)$ and one from its complement.

Check the remaining inequalities (8), then canonicalize the full subset under D_5 . The choice of I_0 changes efficiency, not coverage: every admissible B appears in exactly one of the two cases.

This run tests 50,504,405 filtered clone candidates. It accepts 200 pairs at canonical A representatives, which become 199 full-subset orbits. The final deduplication is needed because a stabilizer of A can act nontrivially on B .

A separate implementation constructs all independent sets of M_5 and computes $\alpha(A)$ by a maximum subset-zeta transform, rather than the primary recurrence. It enumerates clone sets through recursive capacity propagation of (8), rather than the anchor-neighborhood filter. It explores 184,588 recursive nodes and independently produces exactly the same sorted 199 canonical masks. Its completeness proof checks every pruning rule: saturation, insufficient remaining vertices, and the residual cardinality bound of a capacity constraint.

Both methods cover every original-set orbit and every clone choice permitted by (8). All 1,990 labelled subsets in the output orbits are checked against all 857 maximal independent sets. Every stabilizer is trivial. The orbit counts in Theorem 1.1 follow.

5 Reproducibility and verification boundary

The core verification commands are below. Run the first in `mycielski_pilot` and the remaining commands in `mycielski_strengthening`. The accompanying manifests identify all source files, certificates, orbit data, and independent audit records.

```
python check_exact_certificate.py
g++ -O3 -std=c++17 enumerate_extremizers.cpp -o enumerate_extremizers
./enumerate_extremizers
python check_strengthening.py
```

The independent audit has its own enumeration and checker. Python checking requires version 3.10 or later and only the standard library; regenerating proposed LP weights additionally uses SciPy. The primary complete enumeration has a safety time limit: a limit exit is explicitly incomplete and must not be treated as a proof.

The general structural module is tested on all 75 simple nonempty base graphs of order at most four, including 33,864 induced-subset recurrence checks. These finite tests supplement the general proofs. All proof certificates and enumeration code are preserved with checksums. No floating-point solver status is a proof premise. No proof-assistant formalization is claimed, and no conclusion about the asymptotic ratio $\chi_f(M_r)/\rho(M_r)$ is inferred.

中文摘要

本文给出第六个 Mycielski 图的 Hall 比的计算机辅助精确证明。采用 $M_2 = K_2$ 、 $M_{r+1} = \mu(M_r)$ 的编号约定, M_6 有 47 个顶点, 且 $\rho(M_6) = 10/3$ 。达到该比值的诱导顶点集恰有 1990 个; 在整个图的自同构群 D_5 作用下, 它们分成 199 个轨道, 每个轨道含 10 个顶点集。每个极值集都有 20 个顶点、独立数 6, 并包含最后加入的顶点。按原顶点、复制顶点和最后顶点的数量划分, 三类轨道数分别为 109、83 和 7。邻域并集恒等式为证书数据提供方便的递归描述, 但不声称带来优于已有通用极大独立集枚举算法的效率改进。对于本例, M_5 的 7407 个独立集生成 M_6 的 857 个极大独立集约束。精确有理数证书证明上界及等号情形的必要限制, 两种不同的整数算法独立完成全部极值集枚举。本文的主要结果是精确 Hall 比和完整的有限极值集分类。本文只陈述可复核的有限结论, 不声称已确立发表优先权, 也不解决该图族的渐近问题。

References

- [1] M. Cropper, A. Gyárfás and J. Lehel, *Hall ratio of the Mycielski graphs*, Discrete Mathematics **306** (2006), 1988–1990. doi:10.1016/j.disc.2005.09.020.
- [2] J. B. Barnett, *The Fractional Chromatic Number and the Hall Ratio*, Ph.D. dissertation, Auburn University, 2016. Auburn dissertation repository, Definition 2.1.1 and Chapter 4.
- [3] R. Steiner, *Fractional Chromatic Number Vs. Hall Ratio*, Combinatorica **45** (2025), article 37. doi:10.1007/s00493-025-00164-0.
- [4] S. Tsukiyama, M. Ide, H. Ariyoshi and I. Shirakawa, *A New Algorithm for Generating All the Maximal Independent Sets*, SIAM Journal on Computing **6**(3) (1977), 505–517. doi:10.1137/0206036.